LIN7076 – Foundations of Computational Linguistics

Linear Regression

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How to pause

Just Five Straight Minutes Of Obama Pauses

What do you think are factors (linguistic, sociological, contextual) influencing how long you pause after you end a sentence?

How to pause

How to **predict** a continuous variable based on **multiple factors**?

- For instance, how to predict the duration of a **pause** at the end of a sentence in a speech, given:
  - the **length** of the sentence (number of words);
  - the **speech rate** of the speaker (in words/minute);
  - the **context** of the speech (formal/informal);
  - the **gender** of the speaker (F/M/other);
  - the **importance** of the topic (unimportant/moderately important/very important).

Plan for today

- Define the **kinds of variables** involved in a linear regression model.
- Define what it means for a function to be **linear**.
- Spell-out how non-numerical variables can be **encoded** and used in a linear model.
- Discuss how a linear model can be **used and interpreted**.
- Explain how a linear model can be **fitted** to the data (time allowing).
- Discuss an improvement of the basic model, called **regularization** (time allowing).

Defining the variables of a linear model

Two main kinds of variable

- The variable we want to predict (e.g. pause duration) is called a **dependent** variable.
- The variables we use to predict it (e.g. sentence length, speech rate, context, gender, topic importance) are called **independent** variables.

A linear regression predicts the **dependent** variable from the **independent** variable(s) assuming their relation is **linear**.

$$\widehat{\texttt{pause}} = f(\texttt{length}, \texttt{speech_rate}, \texttt{context}, \texttt{gender}, \texttt{importance})$$

With f a "linear" function.

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Types of independent variables

- Different independent variables are encoded differently!
- **Quantitative variables** represent "real" amounts that can be combined *via* arithmetic operations (+, - etc.). Two subtypes:
  - **Discrete**, integer-like; e.g. number of words (1, 2, 3, 4...).
  - **Continuous**, e.g. speech rate, pause duration (0, ..., 1.5, 1.6, 1.7, ...).
- **Categorical variables** represent groupings. Three subtypes:
  - **Binary** variables with 2 levels, e.g. a formal vs. informal context.
  - **Nominal**, unordered variables with > 2 levels, e.g. genders.
  - **Ordinal**, ordered variables with > 2 levels, e.g. unimportant, moderately important, and very important topic. The distance between each level may be variable and subjective.

What about... positive/negative sentiment? age in years? socio-economic status (> 1 possibility)? Part-of-Speech (Noun, Verb, Adjective...)?

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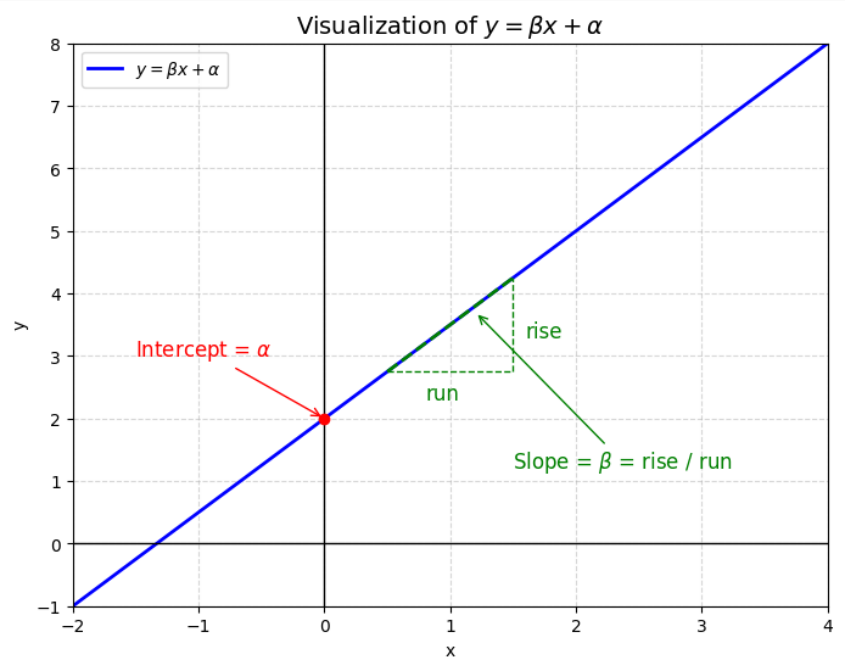
Defining linear relations between variables

Linear relations

A linear regression predicts the dependent variable from the independent variable(s) assuming their relation is **linear**.

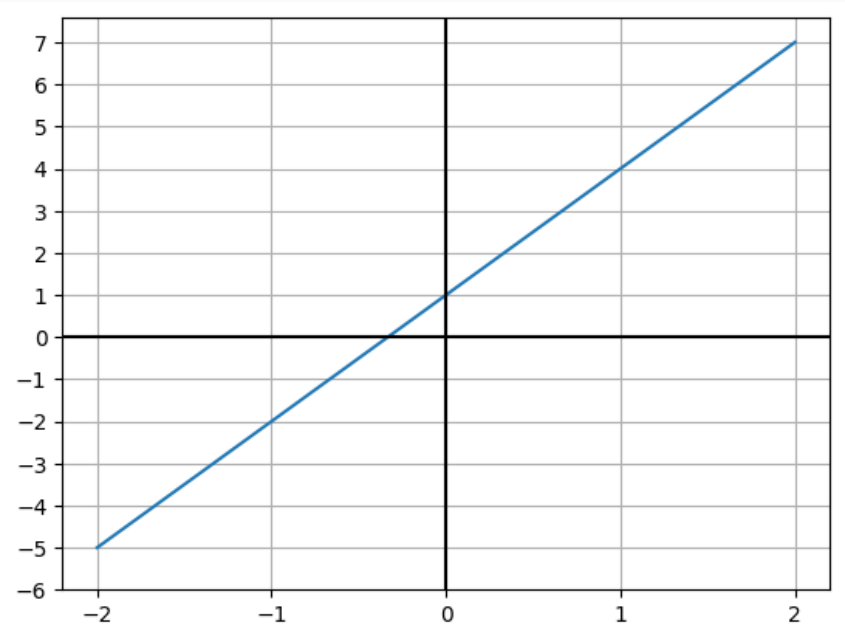
- Two variables x and y are linearly related if changes in x are directly proportional to changes in y at a constant rate β.

$$y = \beta x + \alpha$$

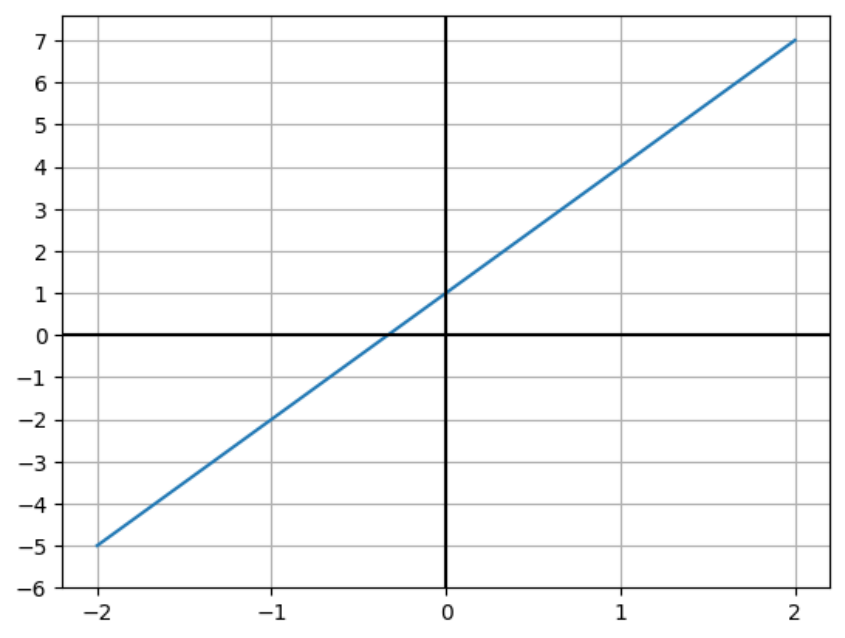


Made with ChatGPT

Linear relation practice 1

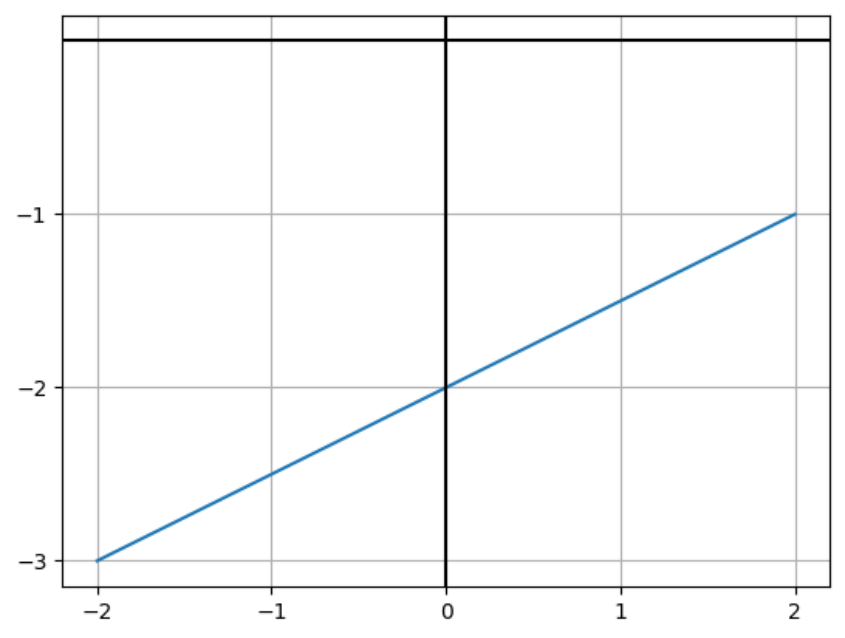


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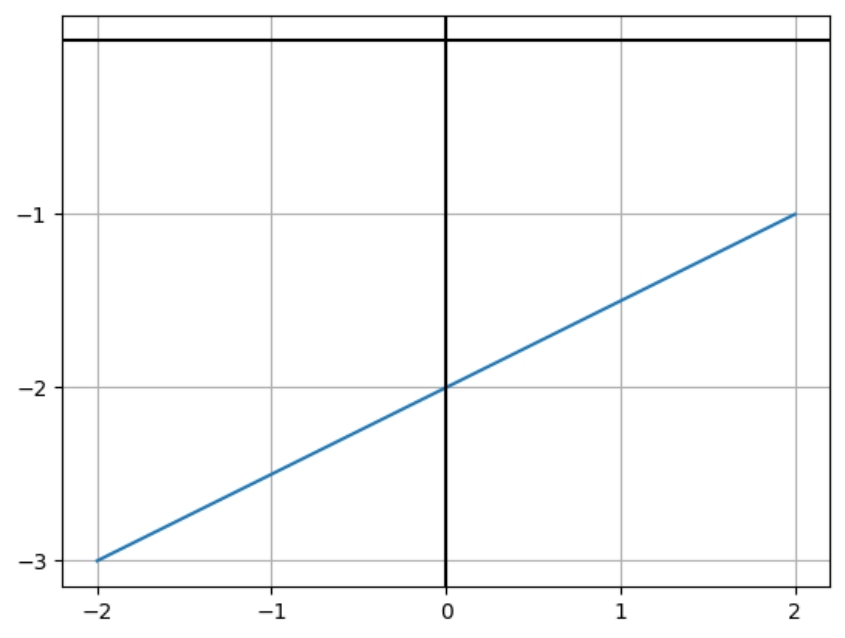


- Curve crosses y-axis at $y = +1$, so intercept α is $+1$;
- Curve goes $+3$ on y for each $+1$ on x: slope β is thus $3/1 = 3$. It's **positive**, consistent with the fact that the curve is **increasing**.
- So the linear function represented here is $y = 3x + 1$.

Linear relation practice 2

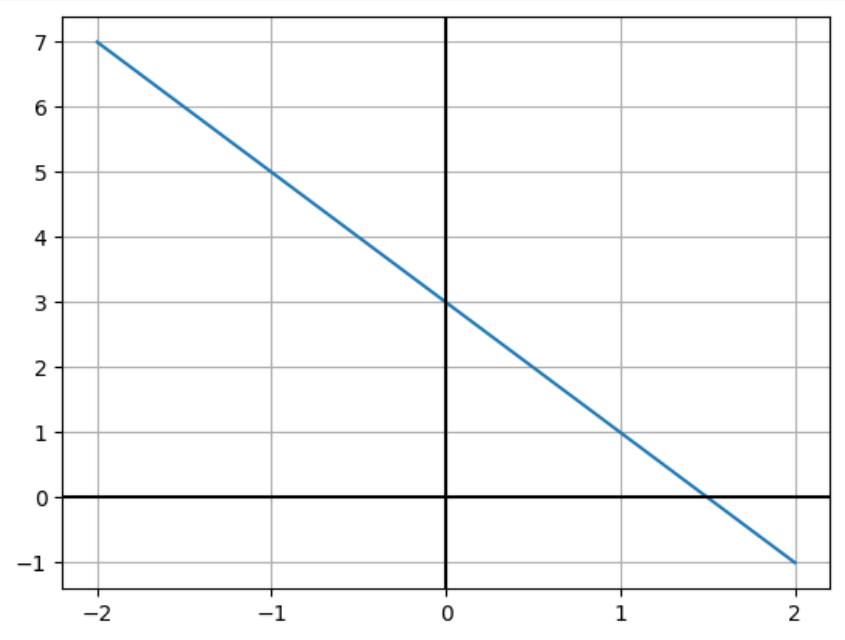


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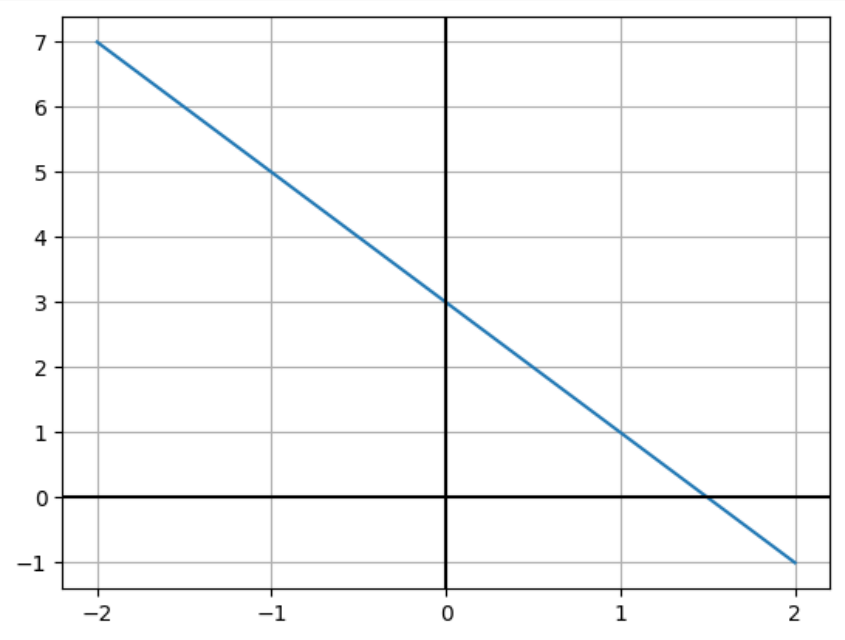


- Curve crosses y-axis at $y = -2$, so intercept α is -2;
- Curve goes $+1$ on y for each $+2$ on x: slope β is thus $1/2 = .5$. It's **positive**, consistent with the fact that the curve is **increasing**.
- So the linear function represented here is $y = .5x - 2$.

Linear relation practice 3



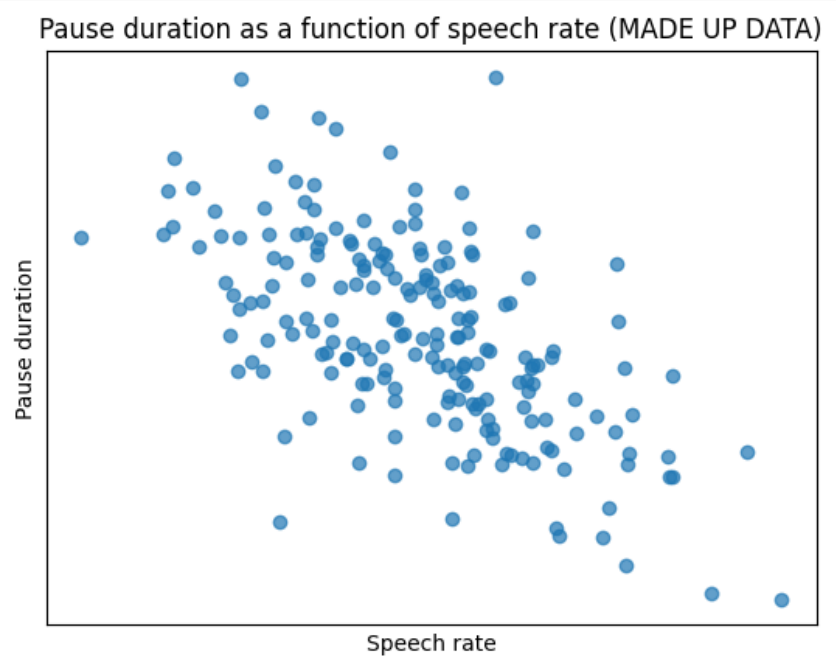
Linear relation practice 3



- Curve crosses y-axis at $y = +3$, so intercept α is $+3$;
- Curve goes -2 on y for each $+1$ on x: slope β is thus $-2/1 = -2$. It's **negative**, consistent with the fact that the curve is **decreasing**.
- So the linear function represented here is $y = -2x + 3$.

Approximating linear relations: negative correlation

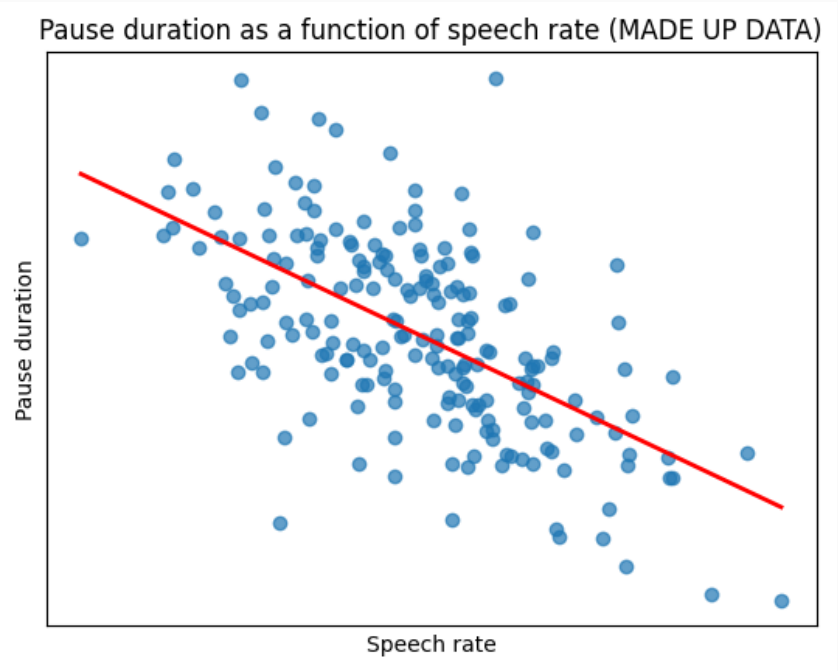
- One can look at **how a single independent variable influences the dependent variable**, e.g. how speech rate (continuous) influences pause duration (continuous).



$$\underbrace{\hat{y}}_{pause} = \beta \underbrace{x}_{rate} + \alpha$$

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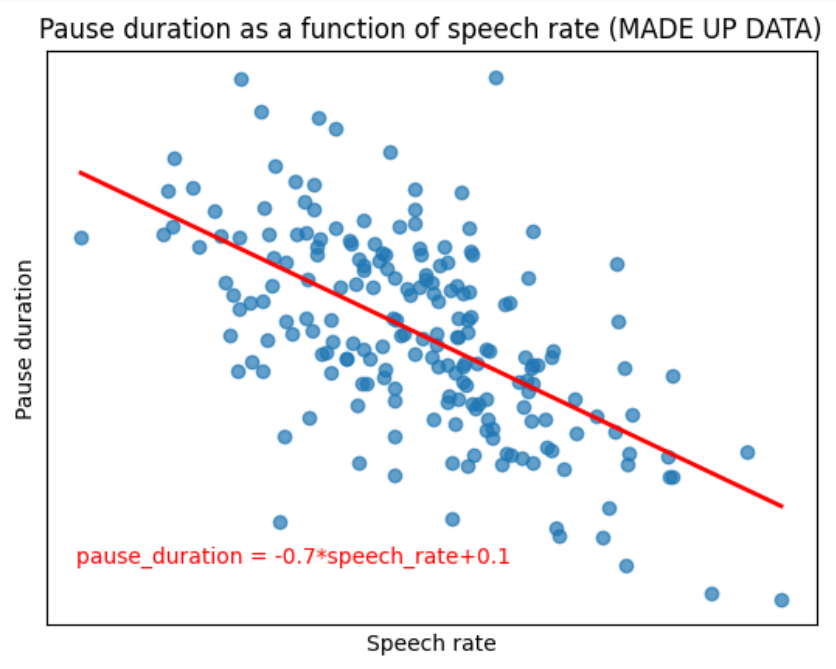


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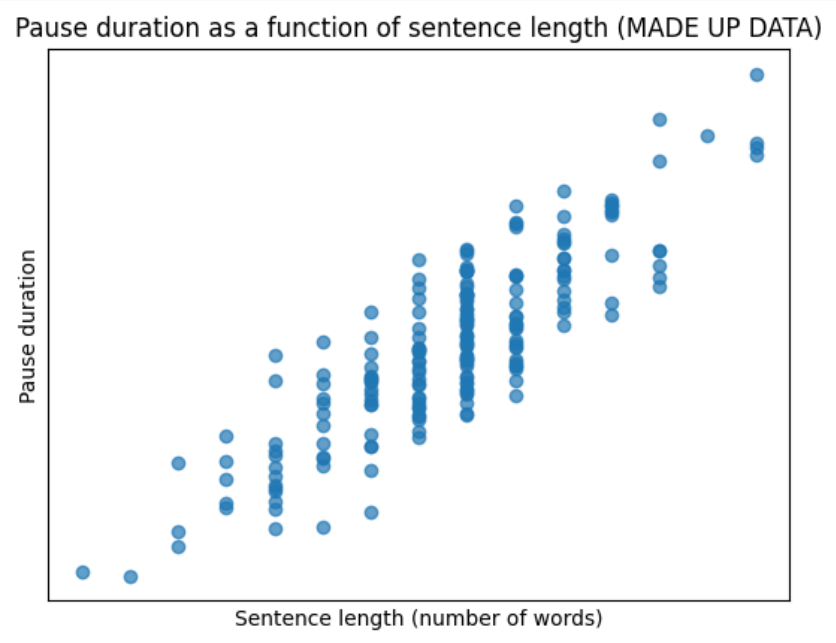


$$\underbrace{\hat{y}}_{pause} = \beta \underbrace{x}_{rate} + \alpha$$

- Sign of β?
- We say that pause duration and speech rate are **negatively correlated**.

Approximating linear relations: positive correlation

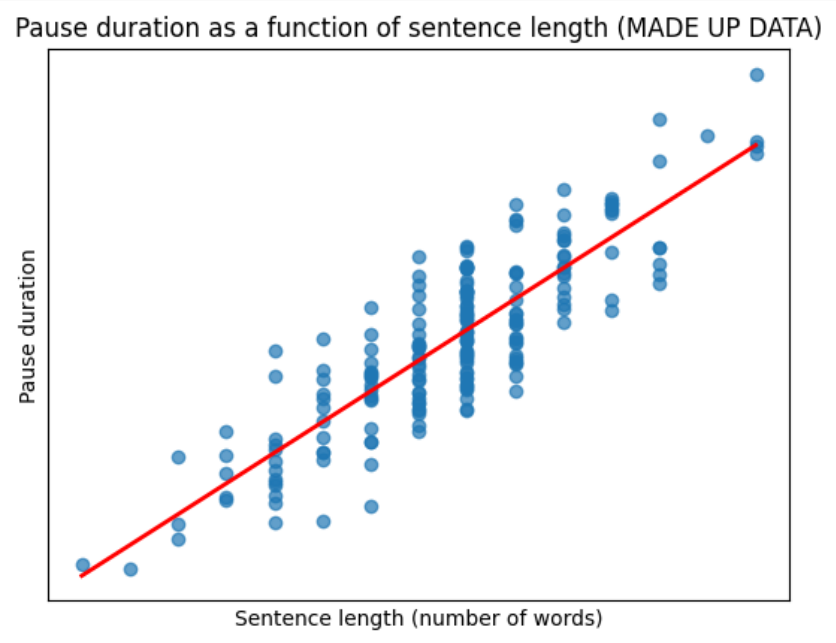
- One can also look at how sentence length (quantitative, discrete) influences pause duration (quantitative, continuous).



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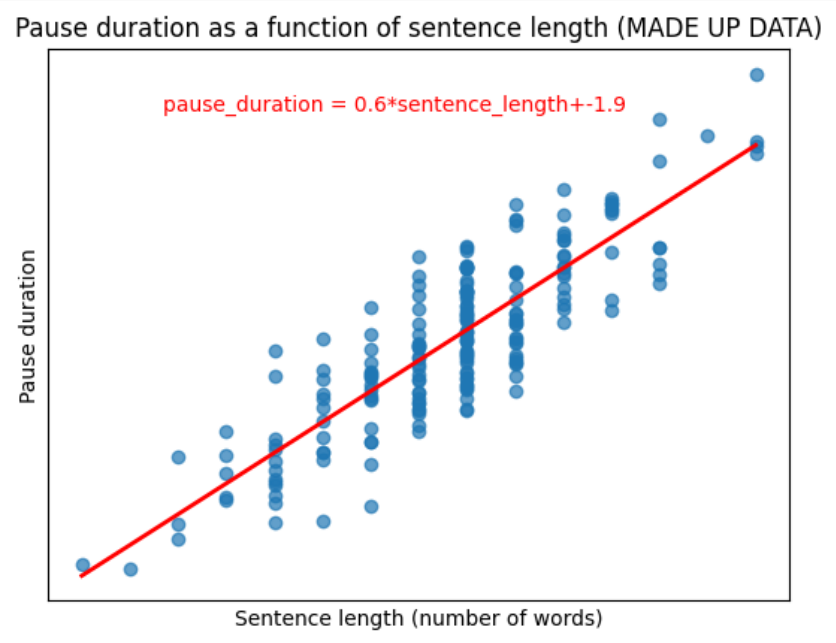


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Approximating linear relations: positive correlation

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$$\underbrace{\hat{y}}_{pause} = \beta \underbrace{x}_{length} + \alpha$$

- Sign of β?
- We say that pause duration and speech rate are **positively correlated**.

Building up the linear regression formula

- We "superpose" several one-to-one linear relations in one single formula. This is still a **linear relation**.

$$\hat{y} = \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n + \alpha$$

- y is a continuous, dependent variable, like pause duration.
- The x_is are the independent variables, and the β_is their respective slopes (**parameters** of the linear model).
  - If variable x_i **positively** influences y, then β_i will be **positive**.
  - If variable x_i **negatively** influences y, then β_i will be **negative**.
  - The **magnitude**/absolute value of β_i ($|\beta_i|$[1]) encodes the **strength** of the association.
- α is the intercept of the model (also a **parameter**). It can be seen as the **baseline value** for y, that would be observed if none of the independent variables were relevant, i.e. if all β_i were zero.

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Sketching a linear model of pause duration

$$\widehat{\texttt{pause}} = \beta_1 \texttt{length} + \beta_2 \texttt{rate} + \beta_3 \texttt{context} + \beta_4 \texttt{gender} + \beta_5 \texttt{importance} + \alpha$$

Sketching a linear model of pause duration

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- Suppose we have a sentence of `length` 15, and the speaker has a speech `rate` or 120 words per minute. We have $\beta_1 = 2/3$, $\beta_2 = -1/10$, $\alpha = 70$ (milliseconds), and all the other βs are 0. What's the estimated `pause` (in milliseconds)?

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$$\widehat{\text{pause}} = 2/3 \times 15 + 1/10 \times 120 + 70 = 10 - 12 + 70 = 68 \text{ milliseconds}$$

What still needs to be fleshed out

$$\widehat{\texttt{pause}} = \beta_1 \texttt{length} + \beta_2 \texttt{rate} + \beta_3 \texttt{context} + \beta_4 \texttt{gender} + \beta_5 \texttt{importance} + \alpha$$

- So far we've seen how to model dependencies between quantitative variables (numbers!), like $\widehat{\texttt{pause}} = f(\texttt{length})$, and $\widehat{\texttt{pause}} = f(\texttt{rate})$.
- What to do with categorical variables which represent groupings, like `context` (assumed binary), `gender` (assumed ternary/nominal), and `importance` (assumed ordinal)?

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Encoding categorical variables

Encoding binary variables

- Binary variables can be encoded as **two different integers**, e.g. 0 and 1. In that case class 0 is the baseline class, and β quantifies the effect of class 1. One could also pick -1 and 1.
- For instance, if we assume the `context` of speech can be informal (c_0) or formal (c_1), the "context" variable will be 0 if the context is informal and 1 if it's formal.

$$\begin{aligned} \texttt{context} &= \begin{cases} 0 \text{ if context is informal} \\ 1 \text{ if context is formal} \end{cases} \\ &= \mathbb{1}_{\text{formal}} \end{aligned}$$

- $\mathbb{1}_{\text{formal}}$ is called the **indicator function** (or characteristic function) for formal contexts.

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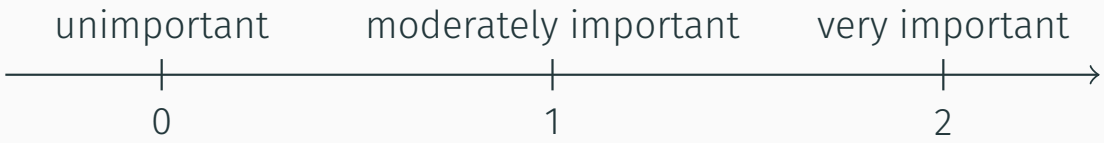
Encoding ordinal variables

- Ordinal variables like topic `importance`, can be **mapped to integers**. This is treating them as discrete.
- For instance, a unimportant/moderately important/very important scale can be encoded as 0/1/2.

- This encoding is assuming that **the "distance" between the different levels is uniform** (because the distance between 0 and 1, and 1 and 2, is the same).
- This is often good enough but if we don't have any clue about how to space the variables, we go for a different encoding called **thermometer encoding** (see last slide of this presentation for more).

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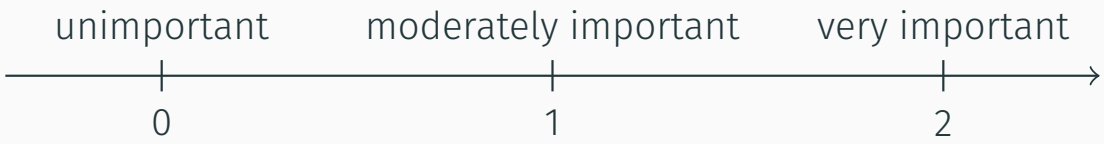
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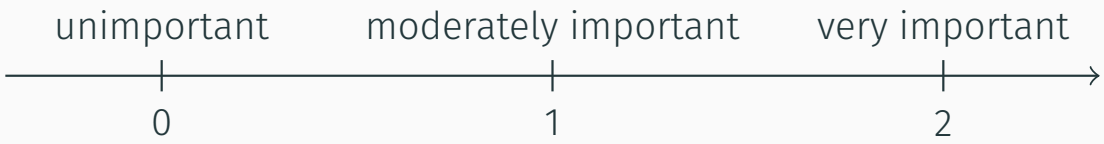
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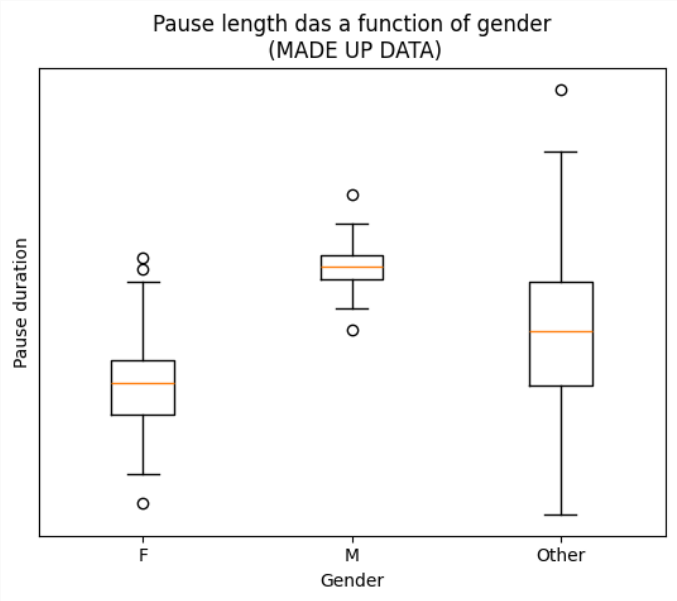


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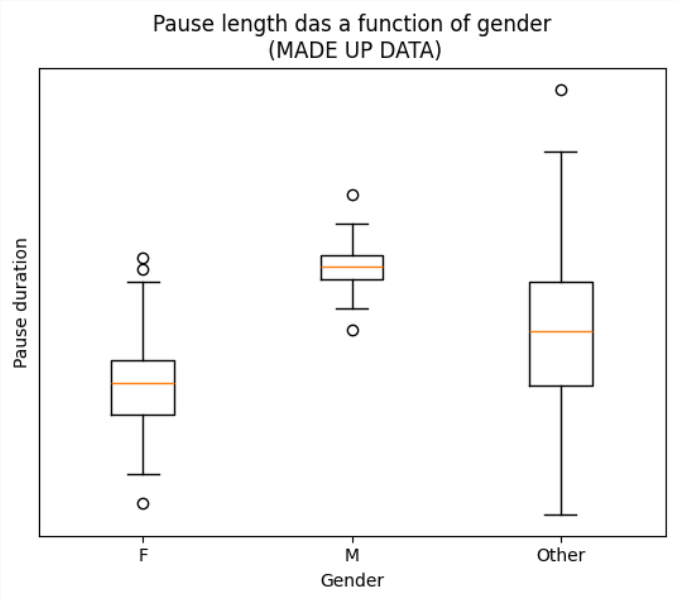


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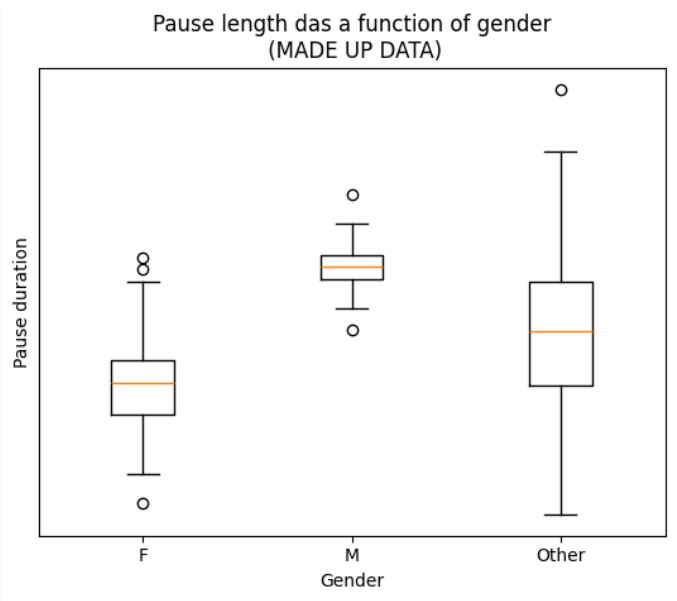


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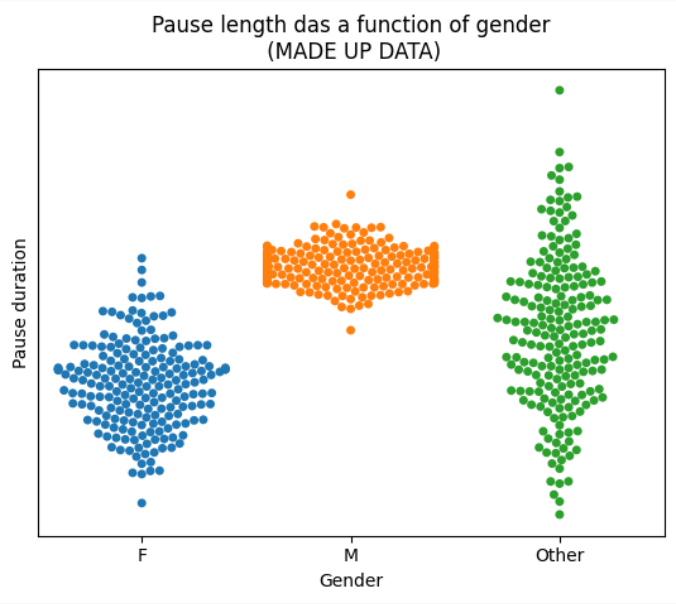
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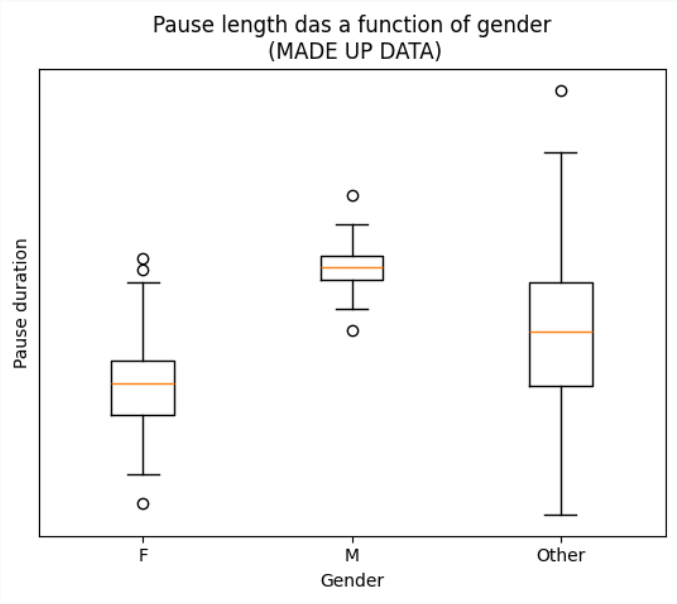




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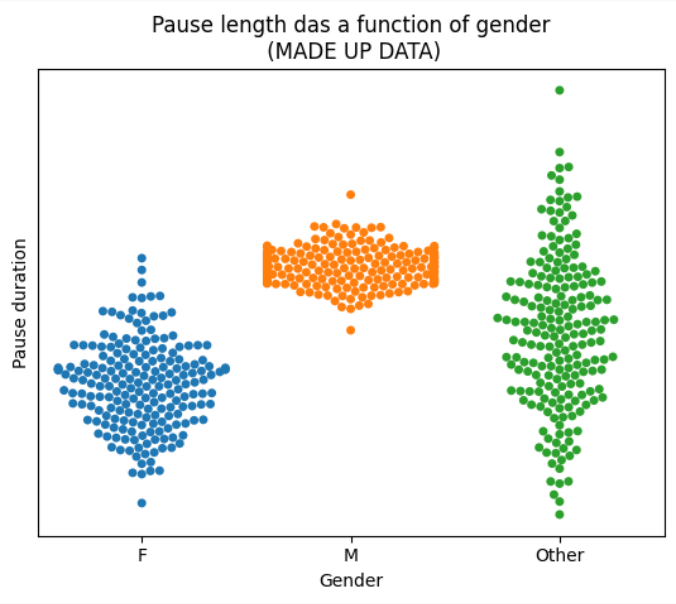
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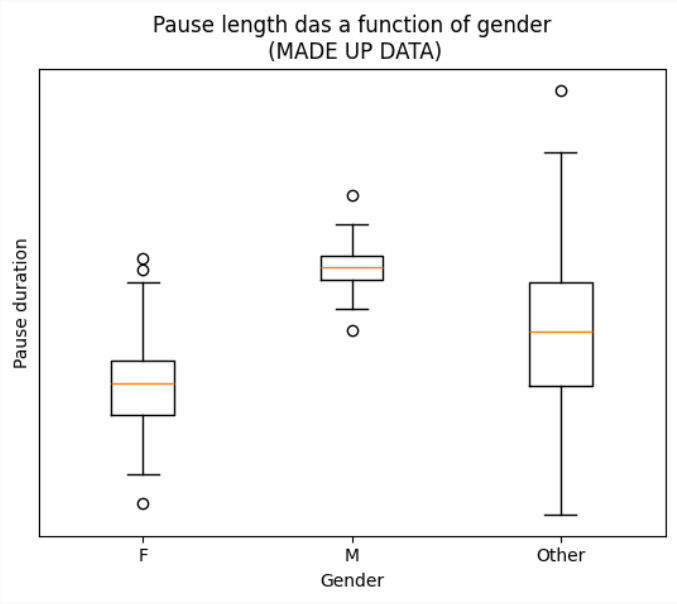




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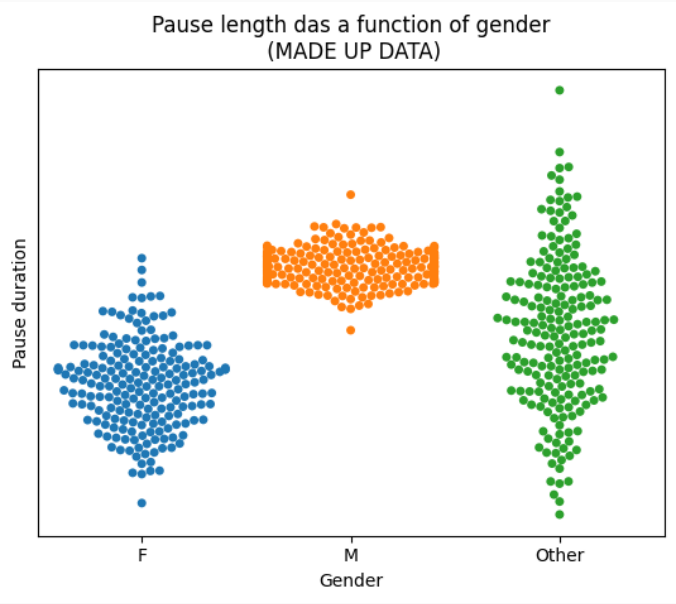
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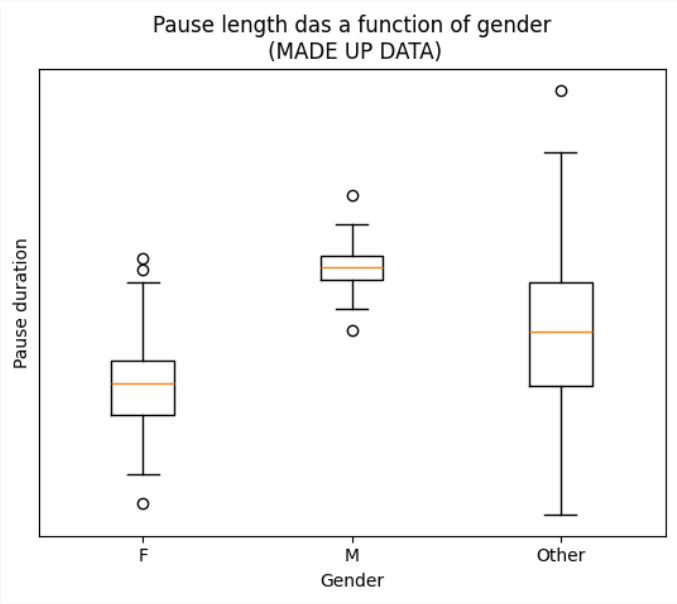




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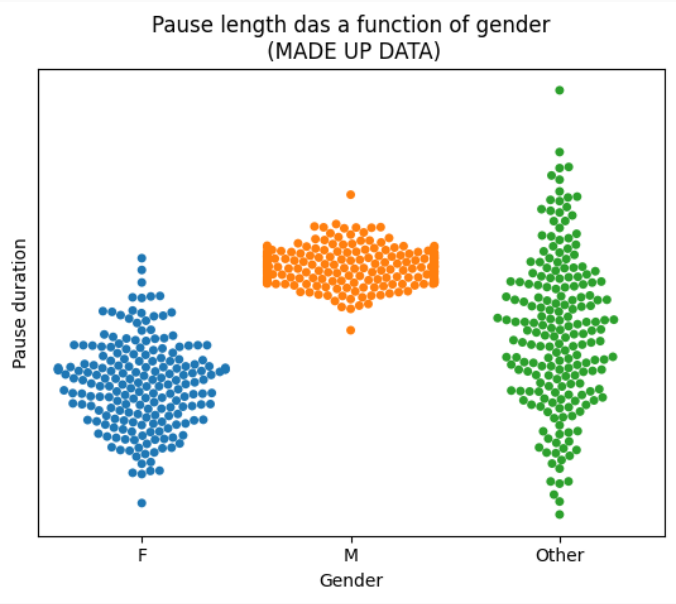
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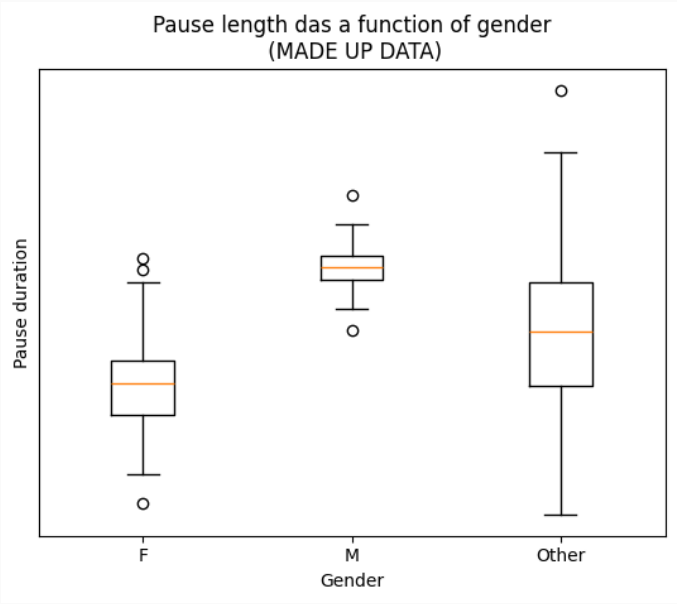




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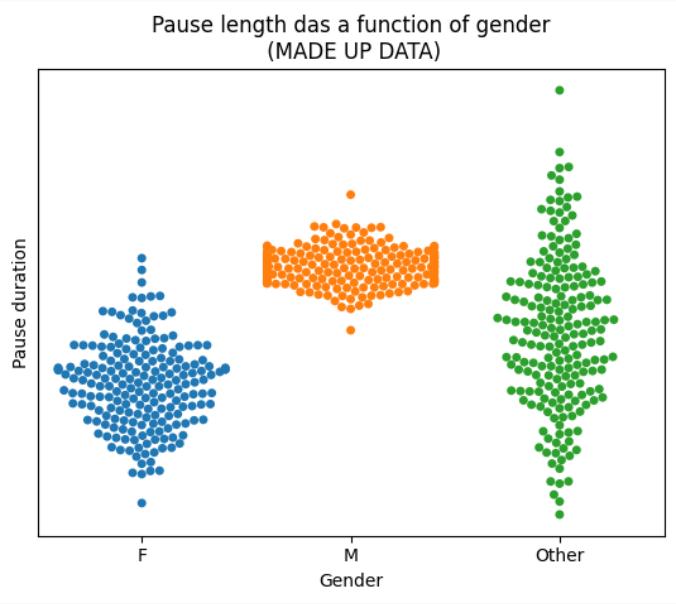
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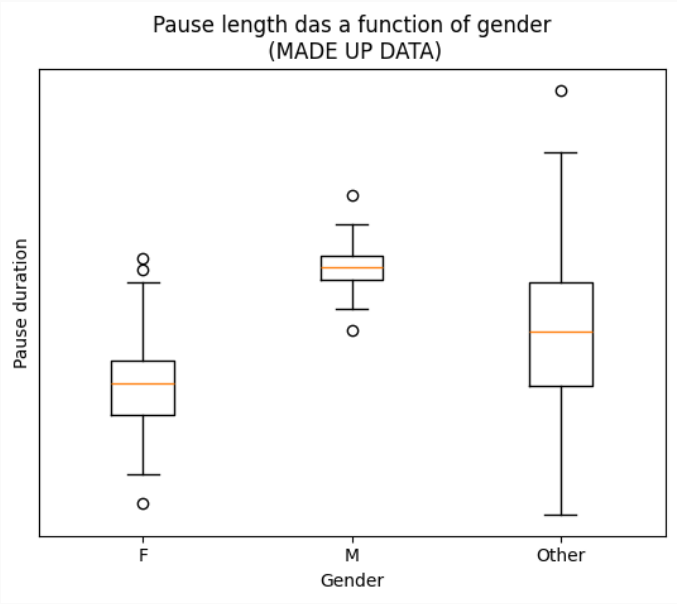




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A full linear model of pause duration

$$\underbrace{\widehat{\texttt{pause}}}_{\text{continuous}} = \beta_1 \underbrace{\texttt{length}}_{\text{discrete}} + \beta_2 \underbrace{\texttt{rate}}_{\text{continuous}} + \beta_3 \underbrace{\texttt{context}}_{\text{binary}} + \beta_4 \underbrace{\texttt{is_male}}_{\text{binary}} + \beta_5 \underbrace{\texttt{is_female}}_{\text{binary}} + \beta_6 \underbrace{\texttt{importance}}_{\text{discrete}} + \alpha$$

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- **Features are standardized to ensure they all vary on the same scale, i.e. they all have a mean of 0 and a standard deviation of 1.** For each feature $k \in \{1, \ldots, n\}$ and input i:

$$x_k^{(i)} \overset{\text{update}}{:=} \frac{x_k^{(i)} - \textsc{Mean}(x_k)}{\textsc{Std}(x_k)}$$

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Using and understanding a fitted regression model

Predictive use

- Linear regression is a **supervised** learning method: we learn the βs and α from known input-output pairs of the form (independent variables, dependent variable).
- Let's assume we did this and so our βs and α reflect the most accurate linear relation between the independent variables and the dependent variable.

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- The **sign and magnitude** of the βs tell us how the independent variables influence the dependent variable!
- Additionally, one can test if any given β is statistically different from 0.[3] This tells us if the corresponding independent variable is a **significant predictor** of the dependent variable–i.e. if it's truly relevant.
- This is useful in **experimental settings**, e.g. when addressing the sociolinguistics question whether gender influences the duration of post-sentential pauses!
- Elaborations of the linear regression model, called **linear mixed effect models**, have become a standard way to test hypotheses in many disciplines, in CogSci and beyond.

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- This is useful in **experimental settings**, e.g. when addressing the sociolinguistics question whether gender influences the duration of post-sentential pauses!
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Hypothesis testing

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Link with neural nets

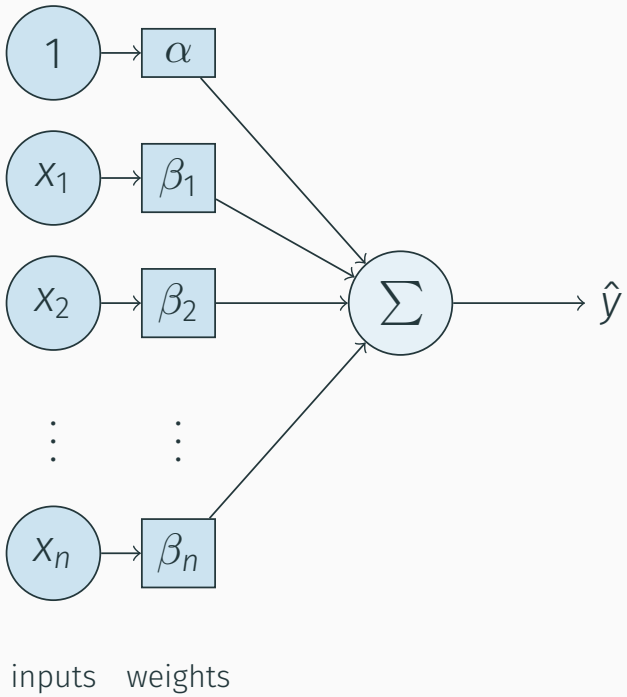
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- Roughly, basic "feed-forward" neural nets are stacked parallelized linear regressions, with some non-linearities added between them.[4]

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Bonus: fitting a linear model

Error for a single input-output pair

- Linear regression is a **supervised** learning method: we learn the βs and α from known input-output (IO) pairs of the form ($\underbrace{(x_1, ..., x_n)}_{\text{independent variables}}$, $\underbrace{y}_{\text{dependent variable}}$). Let's focus on one such pair.
- Intuitively, we want our βs and α to minimize the **difference** between our estimate $\hat{y}$ and the observed value of y.
- What is used in practice (for statistical/analytical reasons), is the **square of the difference** between $\hat{y}$ and y.

$$(\hat{y} - y)^2 = (y - \hat{y})^2 = (\beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n + \alpha - y)^2$$

- "Good" βs and α should lead to a $\hat{y}$ close to the real value y, and $(\hat{y} - y)^2$ should be small.

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The ordinary least square cost function

- In practice, we will fit our linear regression on **many IO pairs**. The i-th IO pair will bear superscript i: $\left(\left(x_1^{(i)}, ..., x_n^{(i)}\right), y^{(i)}\right)$. For a single IO pair, the error is then:

$$\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \left(\beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + ... + \beta_n x_n^{(i)} + \alpha - y^{(i)}\right)^2$$

- We can average this error over all k IO pairs!

$$J(\beta_1, ..., \beta_n, \alpha) = \frac{1}{k} \sum_{i=1}^{k} \left(\beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + ... + \beta_n x_n^{(i)} + \alpha - y^{(i)}\right)^2$$

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Optimizing the OLS cost function

Goal: find $\beta_1, ...\beta_n$ and α that minimize $J(\beta_1, ..., \beta_n, \alpha)$

- With jargon:

$$(\beta_1^*, ..., \beta_n^*, \alpha^*) = \text{argmin}_{(\beta_1,...,\beta_n,\alpha)} J(\beta_1, ..., \beta_n, \alpha)$$

- Luckily, using **squared differences** between $\hat{y}^{(i)}$ and $y^{(i)}$ for all IO pair i provides us with an **analytical** solution to the above optimization problem! Analytical means the solution is given by a formula which is often straightforwardly computable.
- The proof and computation of this exact solution are quite involved though, but luckily we have libraries that do it for us!

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Assessing goodness-of-fit

- A basic linear regression model will fit the βs and α, and return a R^2 (**R-squared**) score.
- The R^2 is between 0 (bad) ad 1 (good) and measures the proportion of the variation in the dependent variable that is predictable from the independent variable(s) – in brief, **how much variance our model explains; how well we fit**.
- Whats a "good" R^2 depends on the discipline.
  - The threshold will be higher (.7-.9) in hard sciences (for which the factors are better known, and precisely measurable);
  - and lower (.3-.5) in social sciences (for which the factors are multifaceted, and sometimes noisy).

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Fixes and improvements

Crazy coefficients

- Suppose we predicting `pause` only based on `context` (formal/informal) and `importance` (unimportant/important):

$$\widehat{\texttt{pause}} = \beta_1 \texttt{context} + \beta_2 \texttt{importance} + \alpha$$

- Suppose that in our training data `context` and `importance` are perfectly correlated: formal contexts always involve important topics, and informal contexts, unimportant topics. We'll train our model on data that look like:

$$\begin{aligned} y^{(1)} &= \beta_1 \times 0 + \beta_2 \times 1 + \alpha = \beta_1 + \alpha && \text{formal/important features} \\ y^{(2)} &= \beta_1 \times 1 + \beta_2 \times 0 + \alpha = \beta_2 + \alpha && \text{informal/unimportant features} \end{aligned}$$

- Issue: if $(\beta_1^*, \beta_2^*, \alpha^*)$ is optimal, so is $(\beta_1^* - 1000, \beta_2^* - 1000, \alpha^* + 1000)$, but these 2 sets of parameters make very different predictions for possible informal/important and formal/unimportant features bundles...

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Regularization

- To prevent the βs from turning crazy when the training data is a little skewed, we add a regularization term to the regression formula, **penalizing high-magnitude βs**.
- This term includes a **hyperparameter** λ, which quantifies how much regularization should influence the prediction.[5]

$$\hat{y} = \underbrace{\beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n + \alpha}_{\text{basic linear regression model}} + \underbrace{\lambda \sum_{j=1}^{n} \beta^2}_{\text{regularization term}}$$

$$\widehat{\texttt{pause}} = \beta_1 \texttt{length} + \beta_2 \texttt{rate} + \beta_3 \texttt{context}$$
$$+\beta_4 \texttt{is_male} + \beta_5 \texttt{is_female} + \beta_6 \texttt{importance} + \alpha$$
$$+\lambda(\beta_1^2 + \beta_2^2 + \beta_3^2 + \beta_4^2 + \beta_5^2 + \beta_6^2)$$

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Advantages of regularization

- Adding a regularization term to the linear model affects the corresponding OLS function! It now includes the exact same regularization term.

$$J(\beta_1, ..., \beta_n, \alpha) = \lambda \sum_{j=1}^{n} \beta^2 + \frac{1}{k} \sum_{i=1}^{k} \left(\beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + ... + \beta_n x_n^{(i)} + \alpha - y^{(i)} \right)^2$$

- To properly minimize this new OLS function, one must keep $\lambda \sum_{j=1}^{n} \beta^2$ low, meaning, one must keep the βs low in magnitude!
- Regularization prevents **overfitting**, in particular a model making extreme predictions for unseen inputs.
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Lab

https://colab.research.google.com/drive/1yXV8ygn32gTABpJhd6z4oq1o1HFcR_k5?usp=sharing

Appendix

Thermometer encoding

- Suppose we have an ordinal variable whose levels are **not evenly distributed**. Perhaps we are interested in how emotionally affected the speaker is, and want to be fine-grained about extremes: affected/very affected/extremely affected.
- In that case, mapping the levels to integers would wrongly suggest differences between levels are the same!
- Instead, we use **a series of indicator functions, that we see as a "thermometer"**. Our functions are $\mathbb{1}_{\leq \text{affected}}$, $\mathbb{1}_{\leq \text{v.affected}}$, $\mathbb{1}_{\leq \text{v.v.affected}}$.
  - To encode an affected mood, we set $\mathbb{1}_{\leq \text{affected}}$ to 1 and the 2 other variables to 0 ($\simeq$low temperature thermometer).
  - To encode a very affected mood, we set $\mathbb{1}_{\leq \text{affected}}$ and $\mathbb{1}_{\leq \text{v.affected}}$ to 1 and $\mathbb{1}_{\leq \text{v.v.affected}}$ to 0 ($\simeq$mid temperature thermometer).
  - To encode an extremely affected mood, we set all variables to 1 ($\simeq$high temperature thermometer)
- The number of variables set to 1 is like the level of a thermometer. But the differences between levels are qualitative instead of quantitative.